Kingdom of Saudi Arabia

Ministry of Education

King Faisal University

College of Computer Sciences and Information Technology

Fundamental Of Software Engineering course project

Section: 63

Madrasati application

Milestone I

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Abstract

In today's computer world, education has changed a lot due to new technology. The Madrasati educational application is an example of this change. The application helps teachers and students communicate easily. This project is about improving the Madrasati application by adding new things to it and studying how it was built. The main goal is to make the Madrasati application work better for teachers and students by adding new ideas and checking if it can handle more users and works well. By listening to users and improving it one by one, this project wants to make Madrasati application better.

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1.Introduction

The rapid spread of information and communication technology has made the internet an essential tool for educational institutions to connect with their students. E-learning is increasingly being used by schools, colleges, and universities as a flexible and accessible method of teaching and learning. It offers numerous advantages over traditional learning methods, such as the ability to be used at any time and from any location.

In response to the COVID-19 pandemic, schools worldwide, including those in Saudi Arabia, had to close and implement remote learning through virtual classrooms. The Ministry of Education in Saudi Arabia created the Madrasati Platform, an online learning management system, to meet the need for online education. This platform became mandatory for schools, requiring teachers to adapt to this new method of instruction. However, concerns have been raised about the quality of education compared to traditional face-to-face instruction. Despite this, the Ministry of Education sees online learning as more than just a temporary solution and considers it a potential strategic choice for Saudi Arabia in the long term.

In response to feedback from students and teachers using the Madrasati application, we will be discussing and improving some aspects of the application properly. It has been noted that certain problems are being faced by users, and the application needs to be enhanced. Suggestions have been made to increase certain features in a suitable way to improve the overall online learning experience. We will carefully consider all feedback and work towards making necessary improvements to ensure that Madrasati continues to be a valuable tool for both students and teachers.

2.Requirements

2.1 User Requirements:

1. Facilitate seamless entry of attendance and grades through a user-friendly interface:

   -The application should have a user-friendly interface that simplifies the process of entering attendance and grades for teachers, making it efficient and easy to use.

2. Provide timely notifications for assignment and test due dates:

   -The app should include a notification system that sends timely reminders to students about assignment and test due dates, helping them stay organized and submit their work on time.

3. Enable students to receive notifications about upcoming events:

   -Students should receive notifications through the app about upcoming events, exams, and important announcements from the school. This keeps them informed and prepared for school activities, enhancing their overall experience.

4. Ensure accessibility across multiple devices:

   -The application should be accessible on multiple devices, including desktops, to ensure a seamless user experience and convenience for students, teachers, and other users.

2.2 System Requirements:

1. Utilize artificial intelligence (AI):

   - By implementing AI, the application can analyze student data and grades to generate valuable insights regarding overall school outcomes.

   - This can help administrators and educators identify areas of improvement, track student progress, and make data-driven decisions to enhance educational outcomes.

2. Incorporate a feature in the app that allows parents to schedule parent-teacher conferences and meetings with school:

   - The application should include a functionality that enables parents to schedule and manage parent-teacher conferences and meetings with school staff.

   - This feature streamlines the process of arranging important discussions, ensuring that parents and teachers can communicate effectively and address any concerns or updates in a timely manner.

3. Implement a secure login system:

   - To protect user data and maintain privacy, the app should incorporate a robust login system with multi-factor authentication.

4. multi-factor authentication:

   - This additional layer of security verifies the user's identity through multiple factors, such as passwords and unique verification codes, reducing the risk of unauthorized access and enhancing the overall security of personal information stored within the app.

5. Develop a version of the application that is compatible with computer systems:

   -To make the app available to more people, it's important to create a version that works on computers. This way, users can access the app not just on their phones but also on their desktops or laptops.

   - This makes it easier for everyone to use the app and gives them more options for accessing it.

2.3 Functional Requirement:

1. The system should use a chatbot:

   -The system should include a chatbot that automatically answers students' questions

   -Improving user experience by providing fast and efficient responses and reducing the need for manual assistance.

2. The system shall provide an electronic tracking of students:

   The capability to electronically track only students with high grades who want to volunteer to explain lessons to needy students under the teacher's supervision, also calculate the number of volunteer hours they contribute.

   - This feature helps monitor and recognize students' efforts in volunteering, promoting a sense of community and service.

3. The calendar feature:

   - The app should include a calendar feature that shows key school events, holidays, and important deadlines.

   - This feature helps students and parents stay organized, plan ahead, and stay informed about upcoming activities and deadlines within the school.

4. Provide summaries and interactive videos:

   - The system should offer written summaries and interactive videos that are accessible to students with hearing problems.

   - This ensures that these students can access and comprehend the educational content effectively, despite their hearing impairments.

5. The messaging feature for communication:

   - The app should enable direct messaging between parents, managers, and school administrators.

   - promoting efficient communication and collaboration while minimizing the reliance on email or phone calls.

6. Receive the notifications:

   - The system should notify the teacher when a student's performance level drops.

- This helps the teacher stay informed about students who may require additional support or intervention, allowing them to address any concerns and provide timely assistance to the student.

7. Detailed and clear instructional videos:

   - The system should design instructional videos that provide comprehensive and user-friendly guidance on using the application.
   - These videos should be clear, detailed, and accessible to all users, ensuring that they can easily understand and navigate the features and functionalities of the application.

2.4 Non-Functional Requirement:

1. Availability:

   provide a quick response time:

   - The chatbot feature within the application should aim to provide rapid responses, ensuring that users receive answers to their inquiries within a short time frame, specifically less than 5 seconds.

2. Security:

   Implement robust security measures:

   - Strong security measures should be implemented to protect user data and maintain privacy.
   - This ensures that user information remains secure and protected from unauthorized access or breaches.

3. Maintainability:

   Regular updates:

   - The application should undergo regular updates and maintenance to address any errors or problems that may arise.
   - This proactive approach ensures that the application remains effective and reliable for users over time.

4. Performance:

   fast loading time:

   - The application should have fast loading times and minimal downtime to provide users with a smooth and uninterrupted experience.

5. Reliability:

   offline capabilities:

   - Offline capabilities should be incorporated into the application to enable users to access specific features and content even when they don't have an internet connection.
   - This ensures business continuity, particularly in areas with limited connectivity or during network outages.

6. Accuracy:

   verify the accuracy:

   -the system shall not allow for volunteered students to share their explanation on the app until the teacher makes sure of the validity of the student's information they have been explained.

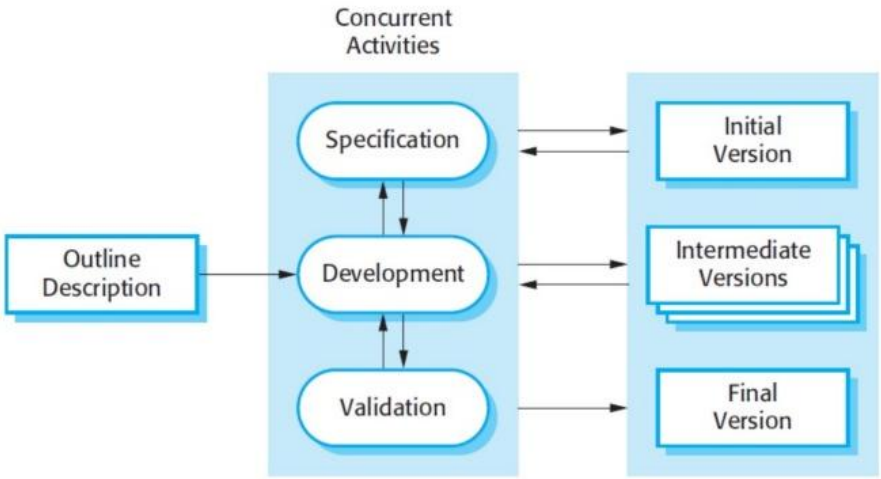
7. Usability:

   Instructional videos:

   -The instructional videos should be brief and designed to occupy minimal storage space when compared to the overall size of the application.

   - This allows users to easily access and view the videos without consuming excessive storage capacity on their devices.

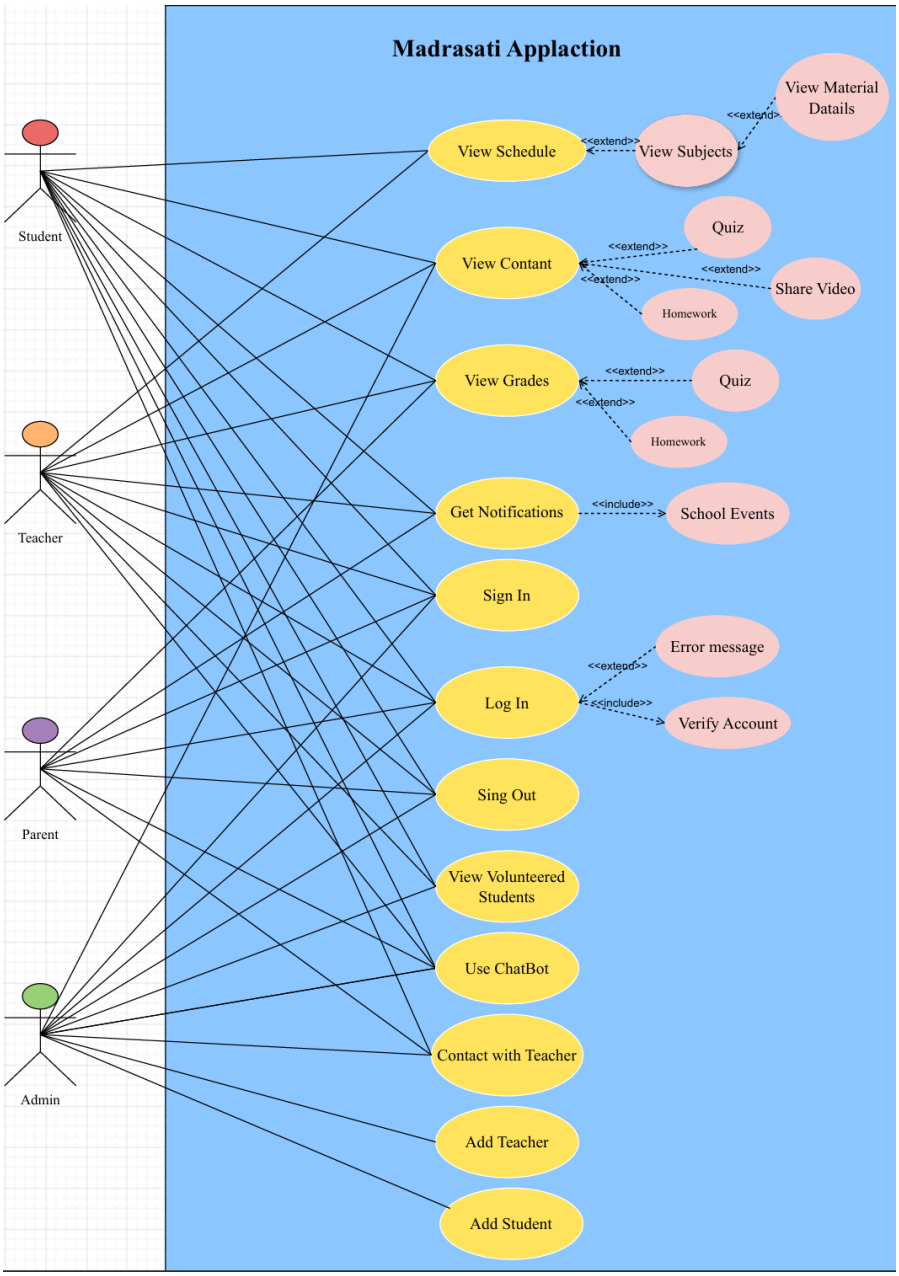
3.Model



Incremental Model – Figure3.1

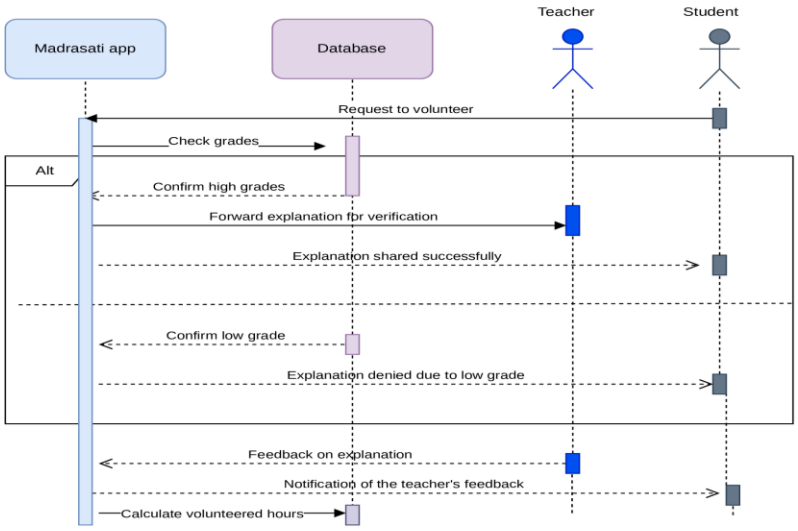
The incremental development model is a good way to develop the Madrasati app. Instead of making the whole app at once, we focus on important parts first. This means users can start using the app sooner and get some benefits early on. With this model, we can easily change things as we go along based on feedback and new ideas. We keep adding new parts to the app regularly. This helps us catch and fix problems early, which saves time and money. We make sure people who are interested in the app are involved in the process, so the app meets their needs. Plus, it helps us manage time and money better by using our resources wisely. To make it work well, we need to plan, work together, and communicate effectively. Overall, using the incremental development model is a good idea for the Madrasati app because it's complex and keeps changing.

4.Use Case Diagram



5. Sequence Diagram

5.1 volunteered students



Sequence Diagram for volunteered students– Figure 5.1

This sequence diagram illustrates the process of a student sharing an explanation with their teacher through an app.

1. The student initiates the process by requesting to volunteer through the app.

2. The app then checks the student's grades in the database.

3. If the grades are high, the database confirms this to the app, which then allows sharing to the teacher.

4. The teacher receives the explanation and reviews it.

5. The app notifies the student that the explanation has been successfully shared with the teacher.

If the grades are not high:

1. The database confirms low grades to the app.

2. The app informs the student that sharing is denied due to low grades.

After receiving feedback from the teacher:

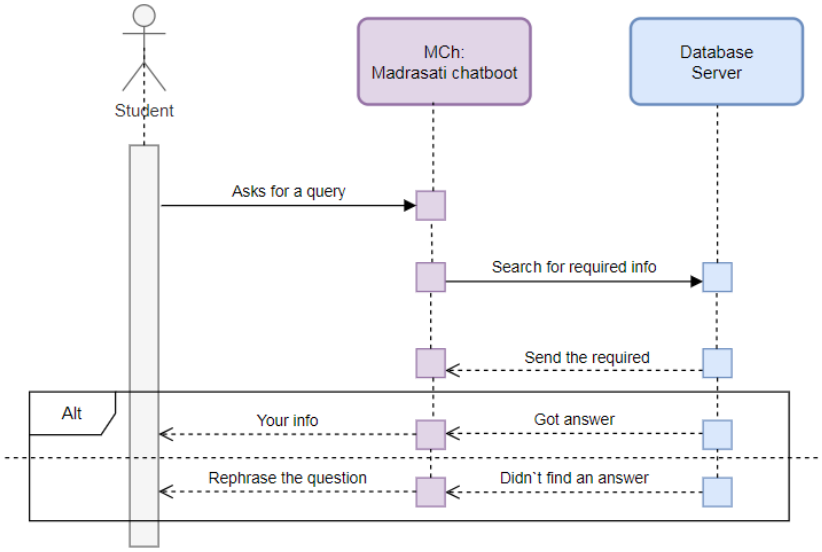
1. The teacher provides feedback on the explanation through the app.

2. The app notifies the student of receiving feedback from their teacher.

3. The app calculates volunteered hours for students.

Overall, this sequence diagram shows a step-by-step process of how a student can share explanations with their teacher through an app and receive feedback on their work.

5.2 The messaging feature for communication



Sequence Diagram for The messaging feature for communication– Figure 5.2

This sequence diagram illustrates the process of using the chatbot feature effectively and getting the best results.

1 .The student begins the process by asking the question in a chatbot through the application.

2 .The application then searches for the answer to the student's question in the database.

3. If the answer is available in the database, the database sends it to the student through the application.

4. If the answer is not obtained from the database, a message is sent to the student stating that the answer was not obtained. Please reformulate the request.

Overall, this sequence diagram shows the step-by-step process of how a student can search for answers through an application and receive model and correct answers for the material.